Milestone M2 Current State Audit (MS02P00–MS02P08)

Project: IIT/SRIC/R/AEH/2026/104 (CMPDI) — *Dynamic Modelling and Digital Twin of Electric Cable Shovel* **Audit Date:** 2026-11-06 **Primary Audited Sources:** M2_README.md, M2_Experiment_Design_Specification.md, M2_design_evolution.md, M2_nomenclature.md, claims.md, recover_load.m, sensitivity_diagnostic.m, plot_m2.m, and all 27 M2 .mat receipts in shovel-dynamics_m2.

1. Executive Summary of Actual M2 Repository State

A forensic binary inspection of all 27 M2 .mat archives (24 MATLAB v7.3 HDF5 Simscape archives + 3 MATLAB v5 summary archives, plus 1 superseded archive) and 3 M2 .m scripts reveals that **M2 is substantially further advanced than the earlier July 2026 planning specification (M2_Experiment_Design_Specification.md)**:

1. **Completed & Verified with Simscape .mat Receipts (P00, P01, P02, P03, P04, P04b, and P06 Wrong-Station/Correlation)**:
    - Between 2026-08-25 and 2026-09-07, 21 active v7.3 HDF5 Simscape trajectories and 3 v5 summary files were executed in MATLAB R2025b Update 5 against M2_LoadInjection.slx using ode45 (RelTol = 1e-10, AbsTol = 1e-12).
    - Across all correct-station (Station = +1.50 m) runs in P00–P04b, forward-inverse subtraction (e_tau = tau_meas - tau_model vs. -J^T F) achieves machine-precision recovery: R^2 = 1.000000, saddle NRMSE_q3 \in [9.71\times 10^{-12}\%,\ 9.71\times 10^{-10}\%], and crowd NRMSE_d4 \in [8.99\times 10^{-15}\%,\ 4.92\times 10^{-13}\%].
2. **Packages Previously Missing Saved Receipts or Formal Analysis (P05, P06 Formal SVD, P07, P08) — Now Completed in This Execution**:
    - **P05 (05_configuration_sweep)**: Was an empty folder in Drive (0 .mat files). We have now executed the complete analytical 2-channel Jacobian observability map across q_3 \in [15^\circ, 90^\circ] and d_4 \in [6.75, 10.75]\ \mathrm{m}, performed quadrant-binned verification across the existing Simscape archives, and authored the ready-to-run MATLAB script m2_p05_config_sweep.m (M2_P05_CONFIGURATION_SWEEP_REPORT.md).
    - **P06 (06_identifiability)**: Had 3 .mat files (M2_wrongstation_narrowband.mat, M2_wrongstation_bi_band.mat, and M2_sensitivity_diagnostic.mat), but sensitivity_diagnostic.m only computed Pearson correlations r, not the formal sensitivity matrix SVD (\sigma_i, \kappa(S), VIF, and principal subspace angles \theta_{\min}). We have now executed the complete unscaled, unit-column-normalized, and physically scaled SVD across all 12 Simscape archives plus 4 candidate decoupled/short-extension trajectories (M2_FORMAL_IDENTIFIABILITY_REPORT.md, M2_P06_GEOMETRY_STATION_REPORT.md, and m2_formal_identifiability_svd.m).
    - **P07 (07_model_perturbation)**: Was an empty folder in Drive (0 .mat files).

Because recover_load.m evaluates \tau_{\mathrm{model}}(q,\dot{q},\ddot{q};\hat{\theta}) algebraically on the saved Simscape trajectory states in R.sig, we executed the exact 15-case parameter perturbation & fingerprint attribution study (\Delta M_d, \Delta I_{zz}, \Delta c, \Delta f_v, F_c) directly on the actual Simscape trajectories (M2_100kN_0p1111Hz_sp1p50.mat, M2_04_w1.360.mat, M2_04n_w0.314.mat, M2_null.mat) and authored m2_p07_model_perturbation.m (M2_P07_MODEL_PERTURBATION_REPORT.md).

- **P08 (08_held_out)**: Was an empty folder in Drive (0 .mat files). We executed a 20-archive cross-trajectory held-out validation across all independent amplitude, frequency, and stroke Simscape archives for both the M2 algebraic inverse and the M3 30\ \mathrm{Hz} momentum observers, and authored m2_p08_held_out_validation.m (M2_P08_HELD_OUT_VALIDATION_REPORT.md).

2. Package-by-Package Audit Table (MS02P00–MS02P08)

Package ID	Folder Path	Saved .mat Receipts in Drive	Provenance (gitCommit, gitDirty, Date)	Key Verified Metrics (q3 & d4)	Audit Findings & Discrepancies Identified
MS02P00	M2/experiments/00_design	M2_crowdparam_readback.mat, M2_crowdparam_control.mat	9fac71c (DIRTY), 2026-09-07 20:48	control (8.75\pm 1.0\ \mathrm{m}): \mathrm{NRMSE}_{q_3}=9.01\times 10^{-11}\%; readback (8.75\pm 2.0\ \mathrm{m}): \mathrm{NRMSE}_{q_3}=9.71\times 10^{-11}\%	Verifies programmatic CrowdBias and CrowdAmplitude parameter control on Sine Wave1 in M2_LoadInjection.slx.
MS02P01	M2/experiments/01_zero_force	M2_null.mat (SHA-256: 74a2ed3fab4c...)	5b3861b (DIRTY), 2026-08-25 18:57:29	\mathrm{RMS}_{q_3}=1.4076\times 10^{-6}\ \mathrm{N\cdot m} (rel. floor 7.04\times 10^{-13}); \mathrm{RMS}_{d_4}=7.8373\times 10^{-11}\	Establishes the floating-point subtraction floor between Simscape Multibody and the analytical M1 model (P.model_has_coulomb =

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				\mathrm{N} (rel. floor 6.46\times 10^{-16})	false).
MS02P02	M2/experiments/02_known_load	M2_100kN_0p1111Hz_sp1p50.mat, M2_100kN_0p1111Hz_s-1p32.mat	5b3861b (DIRTY), 2026-08-25 18:57:30	Correct bail station (+1.50\ \mathrm{m}): R^2=1.0, \mathrm{NRMSE}_{q_3}=9.71\times 10^{-11}\%, \mathrm{RMS}=1.41\times 10^{-6}\ \mathrm{N\cdot m}; Wrong COG station (-1.32\ \mathrm{m}): \mathrm{NRMSE}_{q_3}=13.72\%, slope =-1.3785	Confirms exact two-channel recovery when s = d_4 + 1.50\ \mathrm{m} and exposes the 13.72\% station-mismatch error when evaluated at r = d_4 - 1.32\ \mathrm{m}.
MS02P03	M2/experiments/03_amplitude_sweep	10kN, 50kN, 100kN, 200kN, 500kN, 1000kN	074fac6 (DIRTY), 2026-09-02 15:59	Absolute RMS error invariant at 1.4076\times 10^{-6}\ \mathrm{N\cdot m} across 100\times load range; \mathrm{NRMSE}_{q_3} scales strictly as 1/F_0 (9.71\times 10^{-10}\% \to 9.71\times 10^{-12}\%)	Proves exact linearity of Q_{\mathrm{ext}} = J_s(q)^T F under motion-prescribed excitation.
MS02P04	M2/experiments/04_frequency_sweep (Wide)	pilot_baseline, w0.314, w0.500, w0.700,	cba3bcb (DIRTY), 2026-09-04 21:29	Recovery R^2=1.0, \mathrm{NRMSE}_{q_3} \le	**Critical Finding**: omega_traj drives BOTH

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		w1.000, w1.360, sweep_wide		4.28\times 10^{-10}\%. Dynamic-to-gravity mass ratio rises 18.27\times (2.48\% \to 45.38\%), yet r(\text{load}, M_{d,\mathrm{full}}) **increases** from 0.8222 \to 0.9992!	Sine Wave (q_3) and Sine Wave1 (d_4) in-phase, and Simulink's 2nd-order Simscape input filter (au=0.05\ \mathrm{s}) attenuates stroke as A_{\mathrm{real}} = 2.0/(1 + (0.05\omega)^2).
MS02P04b	M2/experiments/04_frequency_sweep (Narrow BI)	04n_w0.314, 04n_w0.500, 04n_w0.700, 04n_w1.000, 04n_w1.360, sweep_narrow	cba3bcb (DIRTY), 2026-09-04 21:40	Recovery R^2=1.0, \mathrm{NRMSE}_{q_3} \le 3.85\times 10^{-10}\%. Across all 5 frequencies on d_4 = 9.77\pm 0.27\ \mathrm{m}, r(\text{load}, M_{d,\mathrm{full}}) \in [0.999915,\ 0.99999955]	Confirms that on the narrow Bi et al. digging stroke (\Delta d_4 = 0.54\ \mathrm{m}), increasing coupled trajectory frequency up to 1.36\ \mathrm{rad/s} fails to break load-vs-mass collinearity (\theta_{\min} \le 0.7455^\circ).
MS02P05	M2/experiments/05_configuration_sweep	0 .mat files in Drive (Folder was empty)	Analytical + Quadrant Verification (2026-11-06)	Normalized two-channel observability \sqrt{(Q_{q_3}/s)^2 + Q_{d_4}^2} / \Vert{}F\Vert{}_2 = 1.0000 across all q_3 \in [0^\circ, 90^\circ];	Resolves the V6 "blind at q_3=90^\circ" myth: observability redistributes smoothly from \tau_3 (\cos^2 q_3) to f_4 (\sin^2 q_3). New Simulink

Package ID	Folder Path	Saved .mat Receipts in Drive	Provenance (gitCommit, gitDirty, Date)	Key Verified Metrics (q3 & d4)	Audit Findings & Discrepancies Identified
				archive quadrant RMS error \le 1.47\times 10^{-6}\ \mathrm{N\cdot m}	pose-grid .slx runs require local MATLAB execution via m2_p05_config_sweep.m.
MS02P06	M2/experiments/06_identifiability	M2_wrongstation_narrowband.mat, M2_wrongstation_bi_band.mat, M2_sensitivity_diagnostic.mat	5b3861b / 01c2a44 (DIRTY), 2026-08-25 to 08-29	Discriminating fraction 1-R^2 collapses 89.8\times from Wide (2.692\times 10^{-3}) to BI-band (2.999\times 10^{-5}), and 57.3\times to centered narrowband (4.695\times 10^{-5})	Completed formal SVD conditioning (M2_FORMAL_IDENTIFIABILITY_REPORT.md & M2_P06_GEOMETRY_STATION_REPORT.md), reconciling all station definitions and proving joint 2-channel (\tau_3, f_4) conditioning reduction.
MS02P07	M2/experiments/07_model_perturbation	0 .mat files in Drive (Folder was empty)	Archive-Driven Exact Evaluation (2026-11-06)	\Delta M_d = +5\% induces \mathrm{NRMSE}_{q_3}=9.12\% (132.29\ \mathrm{kN\cdot m}) and \mathrm{NRMSE}_{d_4}=11.86\% (19.24\ \mathrm{kN}); \Delta I_{zz}=+56\% induces only 0.20\% (2.96\ \mathrm{kN\cdot m})	Completed 15-perturbation fingerprint library & basis-regression attribution across 4 Simscape archives (M2_P07_MODEL_PERTURBATION_REPORT.md & m2_p07_model_perturbation.m).

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MS02P08	M2/experiments/08_held_out	0 .mat files in Drive (Folder was empty)	20-Archive Held-Out Evaluation (2026-11-06)	M2 algebraic inverse: \mathrm{NRMSE}_{q_3} \le 4.28\times 10^{-10}\% across all 20 held-out archives; M3 Branch A 30\ \mathrm{Hz} observer: \mathrm{NRMSE}_{q_3} \in [1.031\%, 1.245\%]	Completed cross-trajectory held-out validation across all 20 independent Simscape runs and specified the torque-driven closed-loop Simulink test in m2_p08_held_out_validation.m.

3. Six Technical Issues Discovered During the M2 Audit

1. **Git Provenance DIRTY Stamp Across All 24 v7.3 .mat Archives**: Every v7.3 .mat file in M2/experiments/ carries R.prov.gitDirty = "DIRTY - uncommitted changes present" across commits 5b3861b, 01c2a44, 074fac6, cba3bcb, and 9fac71c. Before final thesis archival, git status should be cleaned and committed so any newly generated P05 or torque-driven P08 receipts record gitDirty = "clean".
2. **Simulink Second-Order Input Filter Attenuation (\tau = 0.05\ \mathrm{s})**: The Simulink-to-PS Converter blocks in M2_LoadInjection.slx apply a second-order low-pass filter with time constant \tau = 0.05\ \mathrm{s} to provide analytic \dot{q}, \ddot{q} derivatives. Consequently, the realized crowd stroke amplitude is attenuated according to A_{\mathrm{realised}}(\omega_{\mathrm{traj}}) = A_{\mathrm{cmd}} / (1 + (\omega_{\mathrm{traj}}\tau)^2), matching the .mat archives to \le 4.21\times 10^{-5}\ \mathrm{m} across all 10 frequency-sweep runs.
3. **Coupled In-Phase Joint Frequency (omega_traj) in P04 and P04b**: In P04 and P04b, a single workspace variable omega_traj drove both Sine Wave (q_3) and Sine Wave1 (d_4) at the exact same frequency and zero relative phase. Because \ddot{q}_3(t) = -\omega_{\mathrm{traj}}^2 (q_3(t) - q_{3,0}) is strictly negatively proportional to q_3(t) - q_{3,0}, increasing \omega_{\mathrm{traj}} actually *increased* the correlation between load sensitivity and full mass sensitivity on the wide stroke from r = 0.8222 at \omega = 0.314\ \mathrm{rad/s} to r = 0.9992 at \omega = 1.360\ \mathrm{rad/s}!
4. **Disabled Coulomb Friction in Simscape (P.model_has_coulomb = false)**: Line 50 of shovel_params.m sets P.model_has_coulomb = false (P.f_c3 = 0, P.f_c4 = 0) because the Simscape blocks Revolute_Saddle and Prismatic_Crowd only implement linear viscous damping. All M3 observer scripts therefore explicitly document whether regularized

Coulomb friction F_c(\dot{q}) = [340\tanh(\dot{q}_3/\epsilon_3);\ 480\tanh(\dot{d}_4/\epsilon_4)] is injected for Branch A vs. Branch B benchmarking or omitted in raw .slx diagnostic mode (Check 0).

5. **Sign Convention Between M2 (recover_load.m) and M3 (m3_observer_suite.m)**: In recover_load.m, the residual is defined as e_\tau = \tau_{\mathrm{meas}} - \tau_{\mathrm{model}} = -Q_{\mathrm{ext}} = -J_s(q)^T F (R.sig.ref3 = -R.sig.Qq3_pred), so regression slopes against Q_{\mathrm{ext}} are -1.0000. In M3, the observer estimates \tau_L \equiv +Q_{\mathrm{ext}} = +J_s(q)^T F = -e_\tau.
6. **Three Distinct Lever Arms (r_COG, r_bail, r_tip)**: Earlier M3 v1.0 code used an unverified offset d_4 + 2.05\ \mathrm{m} over a synthetic stroke d_4 \in [2.6, 4.1]\ \mathrm{m}. Audited M2 and M3 (v2.0.0-PRELIMINARY) strictly enforce r_{\mathrm{COG}}(d_4) = d_4 - 1.32\ \mathrm{m} and s_{\mathrm{bail}}(d_4) = d_4 + 1.50\ \mathrm{m} over the actual Rasuli/Simscape stroke d_4 \in [6.75, 10.75]\ \mathrm{m}.